

Valorization of organic wastes through black soldier fly larvae bioconversion: reducing greenhouse gas emissions

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ABSTRACT

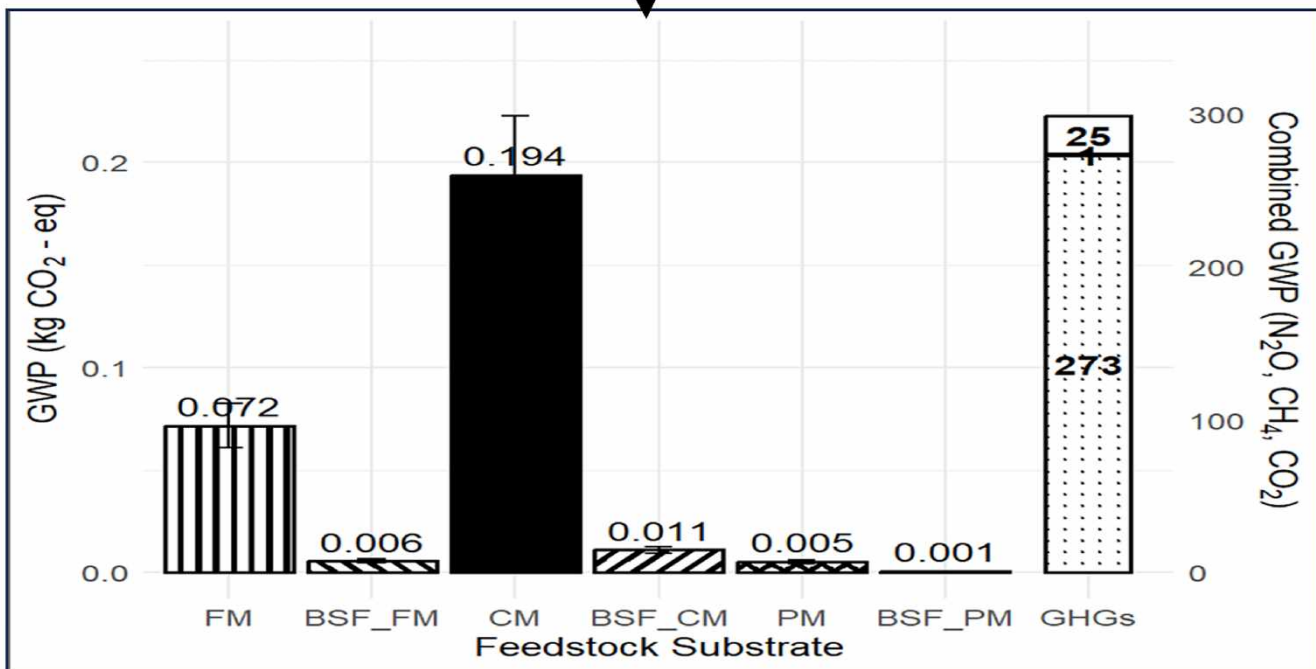
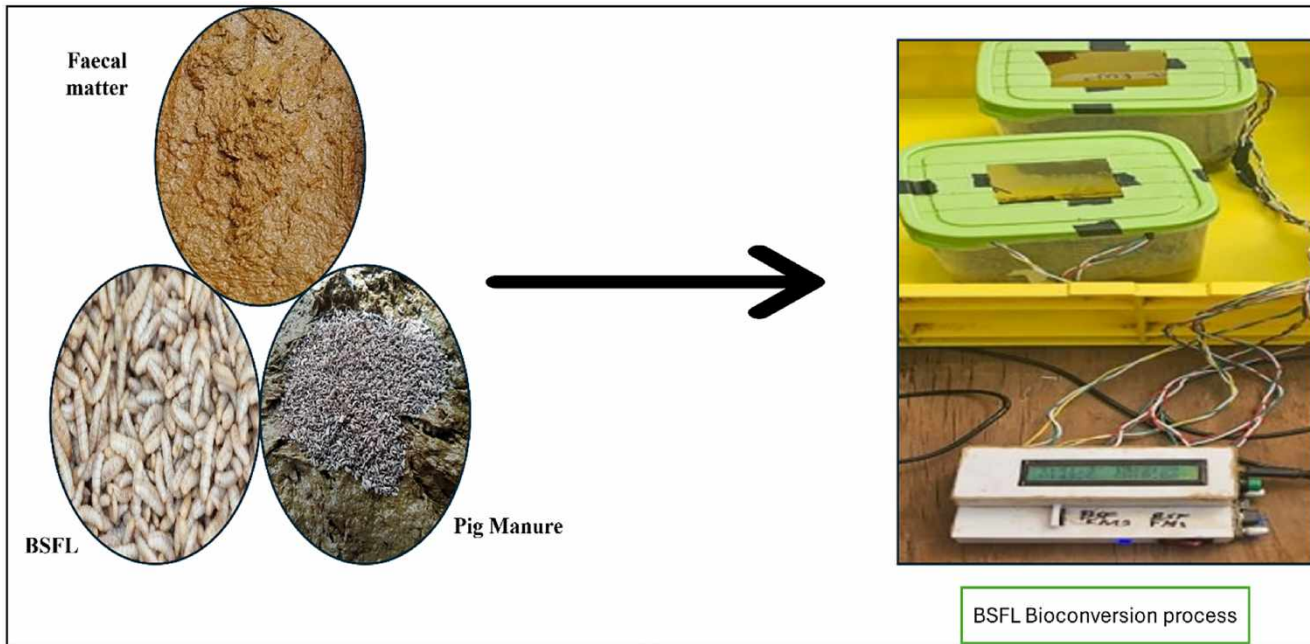
Poor biowaste management contributes to greenhouse gas (GHG) emissions such as methane (CH₄), carbon dioxide (CO₂), and nitrous oxide (N₂O). Black soldier fly larvae (BSFL, *Hermetia illucens*) bioconversion offers a sustainable approach for treating organic wastes while reducing emissions and recovering valuable resources. However, the global warming potential (GWP) associated with BSFL treatment of different waste substrates remains insufficiently characterized. This study quantified GHG emissions (CH₄, CO₂, N₂O, and NH₃) during BSFL bioconversion of human faecal matter (FM), pig manure (PM), and chicken manure (CM) using calibrated gas sensors. Emissions from BSFL-treated substrates were compared with those of untreated controls, and a cradle-to-gate life cycle assessment (LCA) was conducted to determine GWP in CO₂-equivalent (CO₂-eq). In the control treatments, GWP was highest in CM (0.194 kg CO₂-eq), followed by FM (0.072 kg CO₂-eq) and PM (0.005 kg CO₂-eq). BSFL treatment significantly reduced emissions across all substrates, resulting in GWP values of 0.011 kg CO₂-eq for CM, 0.006 kg CO₂-eq for FM, and 0.001 kg CO₂-eq for PM. The greatest reduction was observed in CM, indicating that BSFL treatment was effective for this substrate. These findings demonstrate that BSFL bioconversion can substantially reduce GHG emissions from organic wastes while supporting a circular bioeconomy approach to waste valorization.

Key words: biowaste valorization, black soldier fly, global warming potential, greenhouse gases, life cycle assessment

HIGHLIGHTS

- Black soldier fly larvae (BSFL) mitigate greenhouse gas emissions and ammonia.
- The BSFL treatment was most effective in chicken manure.
- BSFL technology leads to organic waste reduction.

GRAPHICAL ABSTRACT



INTRODUCTION

Poor biowaste management releases potent greenhouse gases (GHG) such as methane (CH₄), carbon dioxide (CO₂), and nitrous oxide (N₂O), which accelerate global warming and drive climate change (Hajam *et al.* 2023). Agriculture accounts for approximately 12% of global greenhouse emissions, equivalent to 5.87 Gt CO₂-eq. Over 65% of these emissions originate

from livestock production, with enteric fermentation as the dominant source contributing approximately 2.8 Gt CO₂-eq, followed by manure management and manure left on pastures (1.2 Gt CO₂-eq) (Symeon *et al.* 2025). The rising global population will intensify meat demand and organic waste generation, amplifying GHG emissions (IISD 2020; Wang *et al.* 2021). Sub-Saharan Africa generates over 62 million tons of organic waste annually, much of which is mismanaged through open dumping and uncontrolled decomposition (Orhorhoro & Oghoghorie 2019). This challenge is more pronounced as rapid urbanization and expanding livestock production in Kenya are expected to markedly increase organic waste by 2030, emphasizing the need for sustainable waste valorization solutions (National Environment Management Authority 2021).

Poor organic waste disposal, often through open dumping or uncontrolled storage, results in the release of greenhouse gases (GHG) such as methane (CH₄), CO₂, and nitrous oxide (N₂O), which have high global warming potential (GWP) and contribute significantly to climate change. These emissions are particularly pronounced in livestock and human wastes due to their high organic matter and nitrogen content, which promote microbial decomposition and gas formation under anaerobic or poorly managed conditions (Amedh 2024). Globally, livestock production generates large quantities of manure, estimated at over 4 billion tonnes per year. In many regions, including developing countries, these wastes are commonly managed through open dumping, lagoon storage, or rudimentary composting systems, which can lead to substantial GHG emissions and environmental pollution (Lema 2025).

Livestock wastes such as pig manure (PM) and chicken manure (CM), together with human faecal matter (FM), are particularly important emission sources during storage, handling, and treatment. PM typically contains high moisture content and readily degradable organic compounds, creating anaerobic conditions that favour methane production. CM, on the other hand, contains high nitrogen concentrations that promote nitrous oxide emissions during decomposition. Similarly, untreated FM can produce both CH₄ and N₂O due to microbial breakdown of organic matter and nitrogen compounds. Composting of livestock manure, although widely used as a management strategy, can still contribute significantly to GHG emissions, accounting for approximately 45–91% of total emissions from manure treatment systems, largely due to release (Nordahl *et al.* 2023), while untreated human FM remains a significant source of CH₄ and N₂O emissions (Lambiasi *et al.* 2024). These characteristics highlight the need for alternative waste management approaches that reduce emissions while enabling resource recovery.

Black soldier fly larvae (BSFL, *Hermetia illucens*) bioconversion has emerged as a promising strategy for transforming organic waste into valuable products while simultaneously reducing GHG and environmental pollution. Unlike anaerobic digestion and composting, which can generate substantial CH₄ and N₂O emissions and require controlled infrastructure (Aguirre-Villegas & Larson 2017), BSFL treatment operates with lower energy requirements and offers rapid waste conversion with nutrient recovery.

However, most existing studies have primarily focused on waste reduction efficiency, larval biomass production, and nutrient recycling, with limited comparative evaluation of greenhouse gas (GHG) dynamics and overall GWP across different waste substrates. In particular, comprehensive assessments of GWP during BSFL bioconversion of FM, PM, and CM remain insufficiently explored (Xiang *et al.* 2024b).

Furthermore, previous studies investigating GHG emissions during waste treatment have largely relied on advanced analytical techniques such as gas chromatography-flame ionization detection (GC-FID), infrared gas analyzers, and differential optical absorption spectroscopy (Lambert-Girard *et al.* 2015; Shellie & Heng 2023). Although these techniques provide high analytical accuracy, their application is often constrained by high operational costs, complex maintenance requirements, and reliance on laboratory-based infrastructure. This limits their suitability for continuous monitoring in small-scale or decentralized BSFL treatment systems (Boakye-Yiadom *et al.* 2022).

Therefore, this study integrates real-time gas sensing using (MQ-4 and MQ-135 sensors with life cycle assessment (LCA) to quantify and compare CO₂, CH₄, N₂O, and NH₃ emissions from human FM, PM, and CM during BSFL bioconversion. By evaluating the relative GWP of these substrates, the study provides insights into the environmental performance of BSFL-based waste treatment systems. GWP expresses the warming effect of different greenhouse gases relative to the emission of 1 tonne of CO₂, thereby providing a standardized metric for comparing the climate impact of different waste management pathways (Shanbhag *et al.* 2024).

MATERIALS AND METHODS

Sampling and preparation

Fresh human FM was collected from a urine diversion dehydration toilet (UDDT) at Gachororo Primary School, Juja, Kenya, while pig and CM were collected from a private farmer in Juja, Kenya. Non-degradable materials were removed, and the substrates were mixed to obtain homogeneous samples. Samples were collected, dried at $104 \pm 1^\circ\text{C}$, and then stored in air-tight containers for elemental analysis. The dried samples were later shipped to Aarhus University in Denmark, Department of Biological and Chemical Engineering, for elemental analysis using vario MACRO cube CHNS elemental analyzer (Elementar Analysensysteme GmbH, Langenselbold, Germany) according to the procedure described by [Muhammad Firdaus *et al.* \(2018\)](#). The elements (carbon, hydrogen, nitrogen, and sulphur) were expressed as a percentage of the total mass of the original sample. Ash content was determined based on an oven dry weight (ODW) basis, which presents the mass of a sample after complete removal of moisture through oven drying as described in Equation (1) ([Feng *et al.* 2005](#)).

$$\% \text{Ash} = \frac{\text{Weight}_{\text{Crucible plus ash}} - \text{Weight}_{\text{Crucible}}}{\text{ODW}_{\text{sample}}} \times 100 \quad (1)$$

The oxygen (%O) was calculated by difference using Equation (2) ([Ellison *et al.* 2023](#)).

$$\% \text{O} = 100 - \% \text{C} - \% \text{H} - \% \text{N} - \% \text{S} - \% \text{Ash} \quad (2)$$

where %C, %H, %N, and %S represent the mass percentages of carbon, hydrogen, nitrogen, and sulphur obtained from CHNS elemental analysis on a dry weight basis.

Sensor calibration and gas measurement

The MQ-4 sensors were used to quantify CH_4 , while the MQ-135 measured CO_2 , N_2O , and NH_3 . The sensors were integrated into a wireless sensor network (WSN) comprising microcontrollers and transceivers for real-time data transmission ([Khalifeh *et al.* 2022](#)). A 72-h burn-in period was conducted to stabilize the sensors, followed by baseline resistance measurements in clean, dry air. Sensor output voltages were measured and monitored across multiple trials to refine the measurement setup. The sensor resistance (R_s) was then computed using the voltage divider (Equation (3)).

$$R_s = \frac{(V_{cc} - V_{out}) * R_L}{V_{out}} \quad (3)$$

where V_{cc} is the supply voltage (5 V); V_{out} is the sensor analog output voltage (volts); and R_L is the load resistance (typically 10k Ω).

The sensor's datasheet was used to derive the gas concentration using the log-log graphs as presented in Equation (4).

$$\text{PPM} = 10 \left(\frac{\log(R_s/R_o) - b}{m} \right) \quad (4)$$

where PPM is the gas concentration in parts per million; m is the slope; b is the intercept; m and b are sensor-specific values obtained from the datasheet for each gas, as shown in [Table 1](#).

Table 1 | Sensor calibration for each gas

Gas concentration	m	b	R_o (k Ω)	R_s (k Ω)
Methane	-2.3	0.72	10	1
Carbon dioxide	-2.2	0.42	30	2
Ammonia	-1.9	0.53	30	2
Nitrous oxide	-1.5	1.05	30	5

Sensor calibration was performed automatically using algorithms, with calibration emissions recorded every 3 min. The control layer transferred real-time data to 'Thinkspeak', an open-source internet platform that receives, processes, and stores the data. The gas concentration was measured at 30 min interval for 15 days.

These sensors were selected due to their affordability, accessibility, and suitability for detecting emission trends in resource-limited research contexts. However, the MQ sensor series is not a laboratory-grade instrument and therefore has limitations in precision, cross-sensitivity, and susceptibility to environmental conditions. These limitations were mitigated through calibration and repeated calibration and trials. The results were interpreted as indicative of concentration trends rather than absolute values. While sensor uncertainty may have influenced the magnitude of calculated emissions, its impact on the estimation of GWP was proportionally consistent across substrates, as all treatments were monitored under the same conditions.

Bioconversion experimental setup

The experiment followed a randomized design in a greenhouse at Jomo Kenyatta University of Agriculture and Technology (JKUAT), Nairobi, Kenya, with three homogenous mixed substrates (FM, PM, and CM). Each substrate was tested with and without BSFL (control), each in triplicate. The larvae were collected from a colony on day 6 and then starved for 24 h before being introduced to the substrates. The starvation period ensured that their guts were emptied, minimizing the risk of carry-over effects from previous feeds that could otherwise bias experimental results (Shishkov *et al.* 2019). Six-day-old BSFL were sampled by passing them through a 1.2 mm mesh screen (Nyakeri *et al.* 2019). Each treatment unit consisted of 250 pieces of 7-day-old larvae, which were fed on 0.25 kg of each substrate according to a procedure outlined by Vodounnou *et al.* (2024). Calibrated sensors were then installed directly inside the ventilated substrate container (115 × 115 × 140 mm) in each treatment unit. DHT22 sensors were used to measure the temperature range (25.94–29.76 °C) and relative humidity level (42.93–66.72%) within the greenhouse.

LCA for BSFL during the bioconversion process

This study used the cradle-to-gate approach, which provides a holistic view of BSFL environmental impacts throughout its entire life cycle. The life cycle inventory (LCI) considered inputs (FM, PM, CM water, chick mash, BSFL eggs, and BSFL), processes (larval rearing and bioconversion), outputs (gaseous emissions (CH₄, CO₂, N₂O, and NH₃) and frass), as shown in Figure 1. Infrastructure construction, chick mash production, and downstream frass utilization were excluded due to negligible contribution to system emissions. Foreground data were directly obtained from experimental measurements of gas emissions (CH₄, CO₂, N₂O, NH₃), while background data were sourced from the ecoinvent database in OpenLCA LCIA methods 2.4.0. The functional unit was the treatment of 0.25 kg of each substrate. The GWP associated with each treatment was subsequently quantified using OpenLCA, an open-source LCA software.

System boundary

The cumulative gas concentration of CH₄, CO₂, N₂O, and NH₃ measured in ppm was converted to mg/kg as per Equation (5) as described by Oyatokun (2023) under standard temperature and pressure conditions (25 °C, 1 atm) to standardize gas concentration across treatments. This conversion assumes ideal gas behaviour, serving as a comparative indicator of emission levels rather than absolute flux.

$$1 \text{ ppm} = \text{mg/kg} \quad (5)$$

Data analysis

The total cumulative emissions at the end of day 15 were presented in a tabular format, as they represented overall trends during the bioconversion process. To enhance reliability, emission values from each treatment were averaged across replicates to allow meaningful comparisons across substrates. The statistical analysis of cumulative gas recorded in ppm for each substrate was conducted in RStudio (version R-4.3.3) using the ggplot2 package for graphical representation. A one-way ANOVA (Analysis of Variance) was performed at a significance level of $\alpha = 0.05$ to test whether mean emissions differed significantly among the control and treated substrates, assuming that the data met the requirements of normality and homogeneity of variances. The carbon-to-nitrogen (C/N) ratio was calculated from the elemental analysis data using the measured %C and %N values. The C/N ratio is a key parameter influencing microbial activity, nitrogen mineralization,

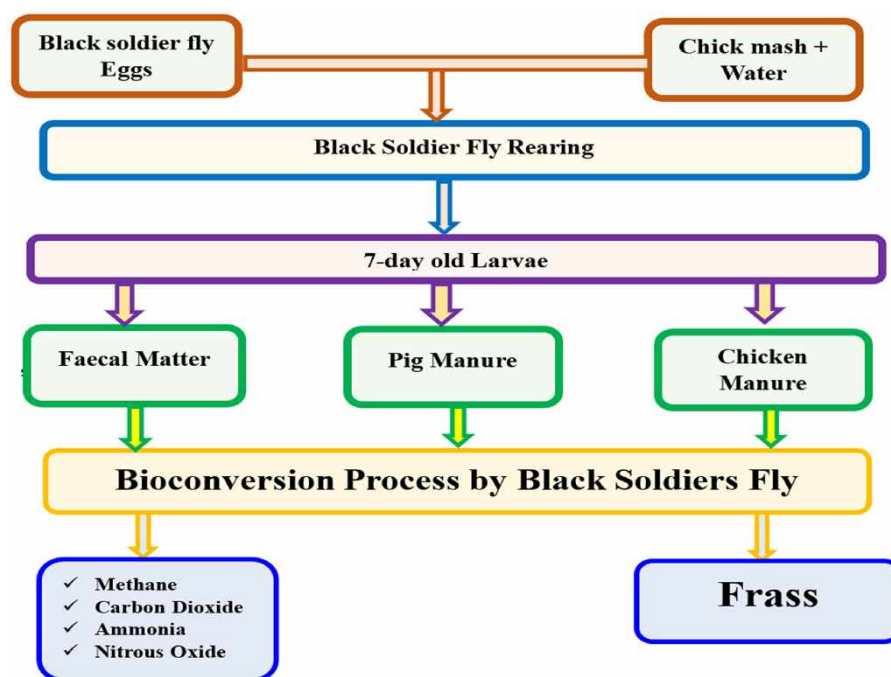


Figure 1 | System boundary (inputs (FM, PM, CM, water, chick mash, BSFL eggs, and BSFL) and outputs (CH₄, CO₂, N₂O, NH₃, and Frass)).

and GHG emissions during organic waste decomposition. Substrates with low C/N ratios tend to favour N₂O emissions due to excess nitrogen availability. LCA was conducted using OpenLCA software (version 2.4) and the life cycle impact assessment (LCIA) method package v2.4.0. The GWP100 methodology was applied to quantify GWP from the measured data, based on the 100-year time horizon outlined in the 2021 IPCC guidelines, where the emitted mass of each gas was multiplied by its respective GWP values (CH₄ = 27.2, CO₂ = 1, N₂O = 273). The software then summed these weighted emissions to derive the GWP, expressed as kg CO₂-equivalent for each substrate.

RESULTS AND DISCUSSION

Temperature and humidity recorded

The recorded temperature exhibited relatively small fluctuations, with a mean of 27.75 ± 2 °C throughout a 15-day period. In contrast, relative humidity displayed more pronounced variability, ranging between $60.72 \pm 4\%$, as shown in Figure 2.

The recorded temperature (27.75 ± 2 °C) and relative humidity ($60.72 \pm 4\%$) fall within the optimal range reported for BSFL development (25–30 °C and 60–70% RH), supporting active larval metabolism and microbial degradation. These conditions are considered favourable for efficient substrate conversion and consistent emission monitoring. Therefore, the environmental conditions during the experiment were suitable for effective bioconversion. These findings highlight the critical role of environmental conditions in regulating BSFL bioconversion and GHG emissions, suggesting that future studies should investigate these effects under controlled environments.

Characteristics of the FM, CM, and PM

Results for the elemental analysis of FM, CM, and PM are presented in Table 2.

Human FM had the highest carbon (43.57%) and nitrogen (5.12%) content, yielding a relatively balanced C/N ratio of ($14.65 \pm 0.11\%$), which supported both microbial decomposition and efficient larval assimilation. This balance facilitated stable nitrification and denitrification processes, contributing to higher CH₄ and N₂O emissions but also sustaining larval growth through sufficient nutrient availability. PM and CM had carbon content of (28.19 and 24.52%) and lower C/N ratios (11.68 and 8.51%), indicating nitrogen-rich substrates prone to NH₃ volatilization, especially in CM with its lowest C/N ratio (Zhou 2016). High ash and oxygen contents in CM (32.31 and 29.10%) further reduced its organic fraction available for larval metabolism, limiting nutrient assimilation and increasing gaseous emissions. FM provided a more

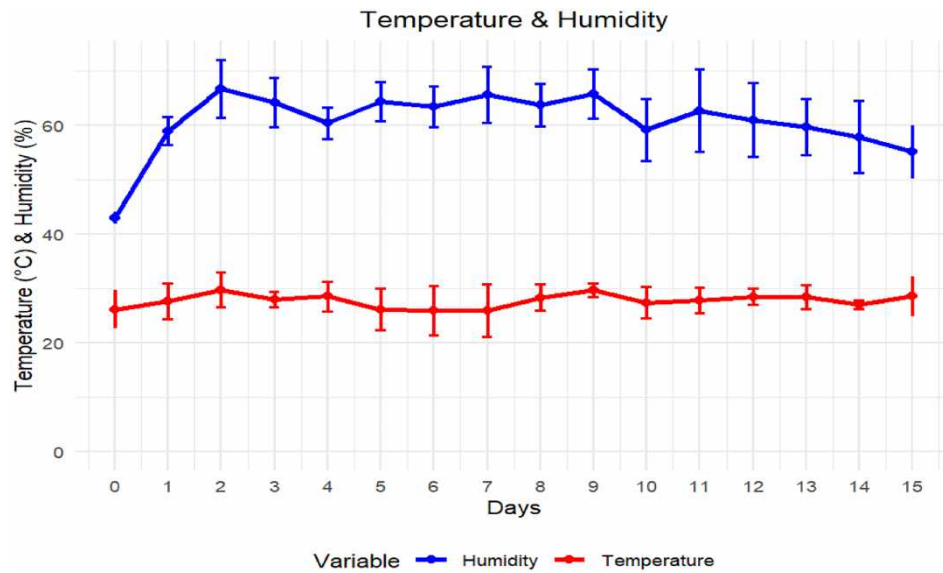


Figure 2 | Temperature and humidity variation during the experiment.

Table 2 | Elemental analysis of the substrates

Feedstock	Percentage weight (dry basis)						
	Carbon	Hydrogen	Sulphur	Nitrogen	Carbon/Nitrogen	Oxygen	Ash
Faecal matter	43.57 ± 0.54	6.51 ± 0.44	0.43 ± 0.12	5.12 ± 0.00	14.65 ± 0.11	14.75 ± 0.02	14.97 ± 0.01
Pig manure	28.19 ± 3.06	4.34 ± 0.49	0.26 ± 0.00	2.41 ± 0.15	11.68 ± 0.51	20.21 ± 0.01	28.91 ± 0.03
Chicken manure	24.52 ± 1.87	3.42 ± 0.23	0.36 ± 0.09	1.68 ± 0.00	8.51 ± 1.21	29.10 ± 0.03	32.31 ± 0.01

Table 3 | Cumulative concentration of greenhouse gases for 15 days

Substrate	Cumulative concentration of greenhouse gases for 15 days (ppm)			
	Methane	Carbon dioxide	Ammonia	Nitrous oxide
BSF_FM	195.75 ± 3.00	195.38 ± 1.00	0.97 ± 0.20	0.10 ± 0.11
FM	2,205.32 ± 160.00	10,210.99 ± 22.00	87.58 ± 58.00	3,619.55 ± 100.00
BSF_PM	20.30 ± 1.29	126.14 ± 1.03	1.22 ± 0.69	0.16 ± 0.60
PM	148.58 ± 0.89	1,176.20 ± 1.70	53.26 ± 0.34	0.55 ± 0.01
BSF_CM	393.57 ± 1.50	138.22 ± 1.80	19.95 ± 0.59	0.83 ± 0.01
CM	3,544.37 ± 74.00	94,853.59 ± 350.00	219.00 ± 1.50	44,294.91 ± 200.00

synergistic nutrient profile for larval growth and microbial activity, while PM and CM presented nitrogen-related imbalances that favoured NH_3 release and constrained bioconversion efficiency. These results were used to explain the contribution of each element in the BSFL bioconversion process of each substrate.

Concentration of greenhouse gases emitted from the bioconversion process

Table 3 presents the cumulative concentration (mean ± SD) of methane (CH_4), CO_2 , ammonia (NH_3), and nitrous oxide (N_2O) measured over 15 days. The results show significant variation across all the gases measured. CM had the highest GHG emissions, with the highest levels of CO_2 (94,853.59 ± 350.00) and N_2O (44,294.91 ± 200.00), reflecting its high organic matter and nitrogen content that promotes microbial decomposition and nitrification–denitrification processes.

FM released moderate concentrations of CH₄ ($2,205.32 \pm 160.00$) and N₂O ($3,619.55 \pm 100.00$), indicating active anaerobic degradation and substantial nitrogen mineralization. In contrast, PM had the lowest emissions across all gases, with CO₂ ($1,176.20 \pm 1.70$) being the most emitted and N₂O (0.55 ± 0.01) the least emitted, suggesting reduced microbial activity and lower availability of degradable organic substrates. Overall, these differences highlight that the type and magnitude of gaseous emissions during decomposition are determined by substrate characteristics, including organic matter quality, carbon content, and C/N ratio, which in turn regulate microbial processes such as methanogenesis, nitrification, and denitrification. Similarly, a study reported high cumulative GHG emissions during composting from food waste and CM, where CO₂ was 186,127 ppm and 173,900 ppm, respectively, highlighting significant emissions of CO₂, CH₄, and N₂O from different substrates tested (Zeng *et al.* 2023).

The BSFL-treated substrates recorded the lowest emissions across all the substrates tested. For instance, BSF_CM emitted 393.57 ± 1.50 ppm CH₄ and 0.83 ± 0.01 ppm N₂O, substantially lower than untreated CM. Similarly, BSF_FM and BSF_PM showed drastic CO₂, CH₄, NH₃, and N₂O emissions compared to FM and PM, respectively, as shown in Table 3. The substantially lower emissions from BSFL-treated substrates indicate that the larvae rapidly consume and stabilize the organic matter, reducing the availability of carbon and nitrogen for microbial processes such as methanogenesis, nitrification, and denitrification. Lindberg *et al.* (2022) confirmed that during composting of fruit and vegetable waste, with or without pre-treatment, CH₄ and N₂O emissions were very low (40–1,570 ppm) compared to CO₂ (47,000–147,000 ppm). A study carried out by Xiang *et al.* (2024a) showed the same trend as the GHG emissions CH₄, CO₂, N₂O, and NH₃ were reduced by the larval bioreactor by 62, 87, and 95%, respectively, compared to composting. BSFL treatment significantly mitigated the emissions of CH₄, CO₂, N₂O, and NH₃. The reduction is mainly linked to improved aeration within the substrate and the larval conversion of organic carbon and nitrogen biomass, which limits the microbial pathway responsible for GHG generation. Additionally, BSFL secretes antimicrobial compounds and enzymes that alter the microbial community, suppressing methanogens and denitrifying bacteria responsible for CH₄ and N₂O generation. This combination of aeration, nutrient assimilation, and microbial stabilization of organic matter lowers overall emissions (Li *et al.* 2025). The results demonstrated that BSFL bioconversion significantly reduced GHG emissions from FM, PM, and CM, highlighting its potential as a climate mitigation strategy. CH₄ and N₂O were particularly reduced, while CO₂ and NH₃ emissions also decreased substantially, further supporting the role of BSFL in enhancing environmental sustainability.

Gas emissions from the bioconversion of the three substrates

Methane emissions from the three substrates

Among the control group, PM showed the lowest cumulative CH₄ emissions, while CM showed the highest emissions. Among the BSFL-treated substrates, BSF_CM recorded the highest cumulative CH₄ emissions, while BSF_PM had the lowest emissions, as shown in Figure 3.

Methane emission peaked during the first few days and declined towards the end, reflecting substrate depletion and reduced methanogenic activity. The emissions were highest in untreated CM (3,544.37 ppm), FM (2,205.32 ppm), and PM (148.58 ppm), and were substantially reduced by BSFL treatment to 393.57, 195.75, and 20.30 ppm, respectively, corresponding to reduction rates of 88.90% in CM, 91.13% in FM, and 86.34% in PM, with CM showing the highest mitigation efficiency relative to its initial emission levels. The low C/N ratio of 8:51 in CM promoted rapid microbial degradation of organic matter, enhancing anaerobic conditions that support methane production. As a result, the combined effect of high ammonia levels and a low C/N ratio significantly boosted methane generation in CM, even under BSFL treatment. BSFL treatment enhanced oxygen diffusion and substrate mixing, promoting an aerobic microbial pathway that converts organic carbon to CO₂ instead of CH₄. This shift suppresses methanogenic processes, resulting in lower methane emissions.

Ermolaev *et al.* (2019) observed that BSFL feeding reduces the presence of readily degradable carbon compounds, thereby limiting methanogenic precursors within the substrate. Matos *et al.* (2021) further noted that BSFL digestion converts a portion of organic carbon into larval biomass, thereby reducing the carbon pool available for microbial methane production. Moreover, Symeon *et al.* (2025) emphasized that BSFL activity shifts microbial community composition, suppressing methanogens while enhancing aerobic and facultative microbial populations. These studies predominantly used closed chamber systems with gas chromatography to quantify CH₄ concentrations, ensuring precise and comparative assessment of gas emission during bioconversion. Collectively, all these studies confirm that BSFL bioconversion effectively mitigates methane generation through biochemical stabilization and microbial regulation.

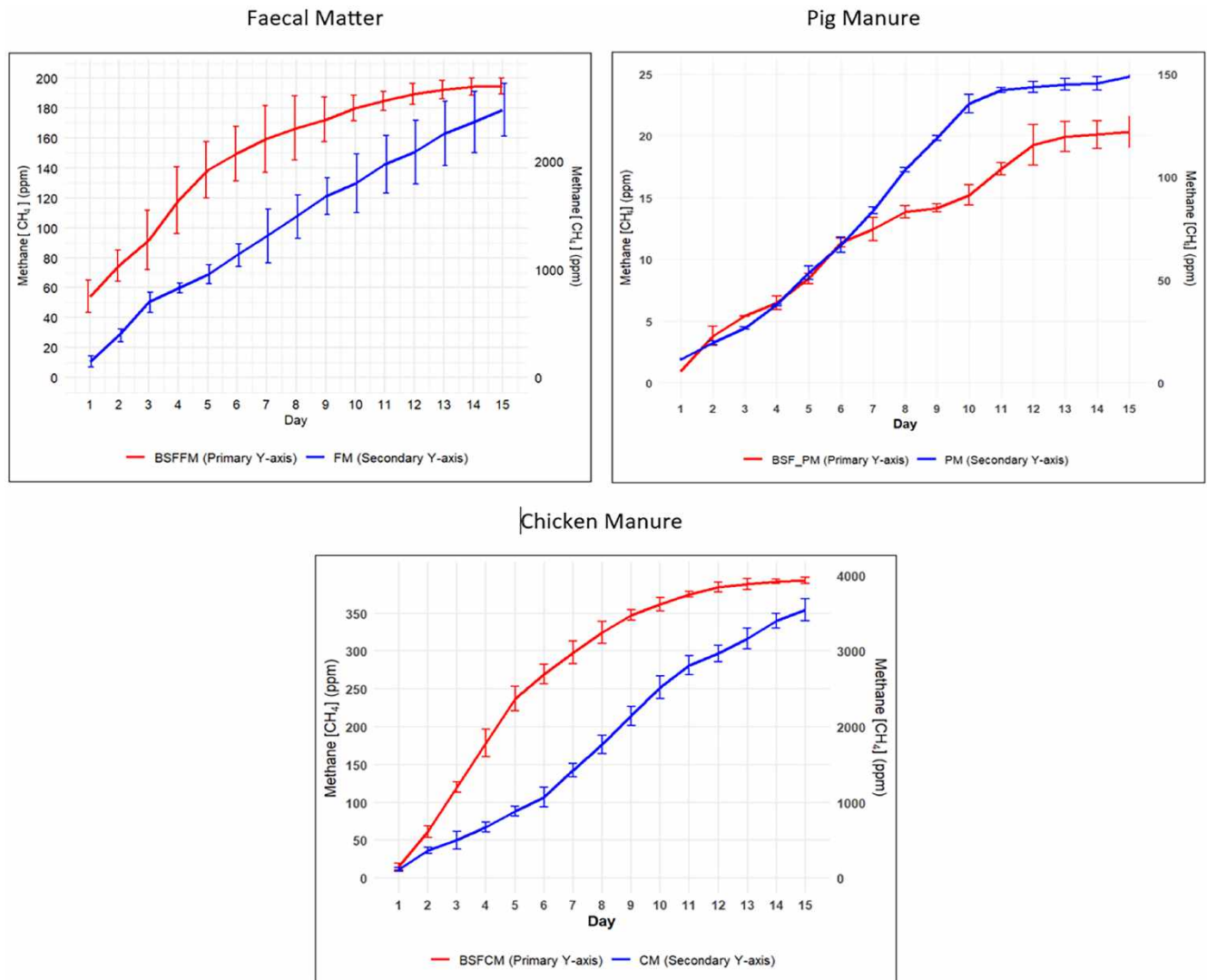


Figure 3 | Cumulative gas concentration for methane emitted from FM, PM, and CM with BSFL and without BSFL.

CO₂ emissions from the three substrates

The cumulative CO₂ concentrations measured over the 15 days are presented in Figure 4.

Initially, CO₂ emissions were low in both treatments during the first few days. However, emissions increased steadily as microbial activity increased, decomposing organic matter in all substrates. Emissions were highest in CM (94,853.59 ppm), followed by FM (10,210.99 ppm) and PM (1,176.20 ppm). BSFL treatment reduced these emissions to (138.22 ppm), (195.38 ppm), and (126.14 ppm), with reduction rates of 99.85, 98.09, and 89.27%, respectively. The emissions were primarily influenced by the carbon content and C/N ratio of the substrate. CM had the highest cumulative CO₂ concentration among the control substrates, followed by FM and PM. FM with high carbon (43.57%) and a balanced C/N ratio (14.65%) supported elevated microbial respiration, whereas CM, despite its lower carbon (24.52%), contained a high proportion of readily degradable organic matter and a low C/N ratio (8.51), which accelerated microbial metabolism and led to high CO₂ release. However, the CO₂ levels recorded for the treated substrate with BSFL suggest that the use of BSFL could contribute to better atmospheric CO₂ regulation and potentially offer an eco-friendly solution in reducing the GHG emissions associated with climate change impact.

BSFL-treated FM generated minimal CO₂ compared to extremely high levels from untreated FM, with one-way ANOVA confirming a difference of ($P = 7.31 \times 10^{-7}$), far below the 0.05 significance threshold, indicating BSFL treatment effectively

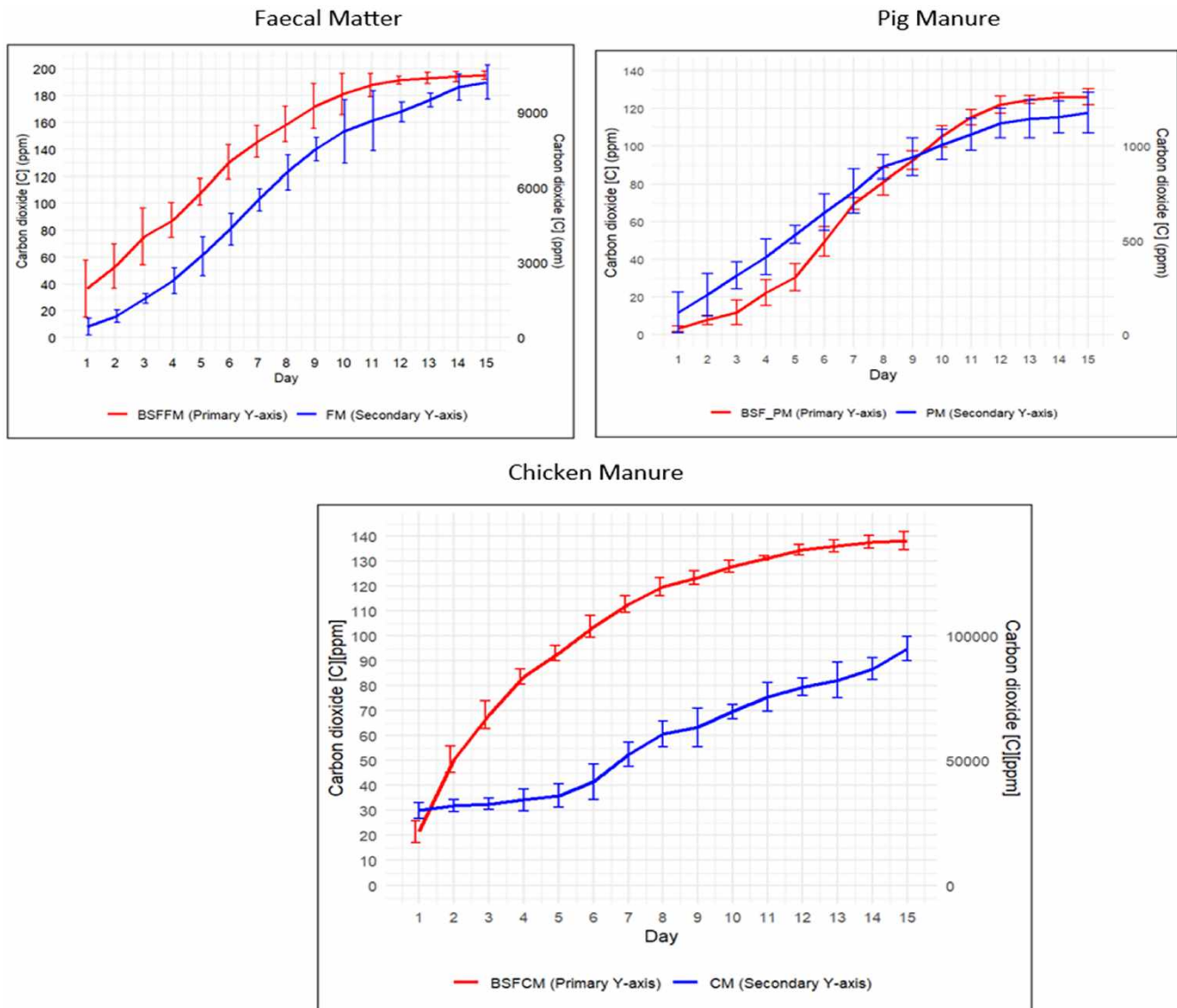


Figure 4 | Cumulative gas concentration for carbon dioxide emitted from FM, PM, and CM with BSF and without BSF.

reduces emissions. The results aligned with findings on container-based sanitation in Nairobi, Kenya, in a study by [Okeny & Niwagaba \(2024\)](#), where the total emission of CO₂ was 400,000 ppm emitted from untreated FM, which was higher than that of BSF_FM. Similar results were observed by [Wu et al. \(2024\)](#), who reported a 32% reduction in carbon loss during BSFL bioconversion on CM; [Cai et al. \(2024\)](#), who demonstrated that BSF stabilizes organic carbon through metabolic assimilation. Additionally, larval activity promotes the formation of humified organic matter and stable carbon fraction in frass, contributing to long-term carbon sequestration. This supports the hypothesis that BSFL may mitigate CO₂ emissions by facilitating aerobic decomposition and limiting microbial respiration. The significant *p*-value (1.19×10^{-10}) further confirms that the BSFL bioconversion process promotes efficient carbon utilization and long-term stabilization within the substrate.

Ammonia emissions on the three substrates

Ammonia emissions varied significantly across all the substrates, as illustrated in [Figure 5](#).

Across all substrates, NH₃ concentrations increased progressively over time, indicating active nitrogen mineralization during decomposition. BSFL-treated substrates generally showed lower NH₃ emissions with a reduction rate of 90.89, 98.89, and 97.71% from CM, FM, and PM, respectively, suggesting nitrogen losses due to larval assimilation and improved

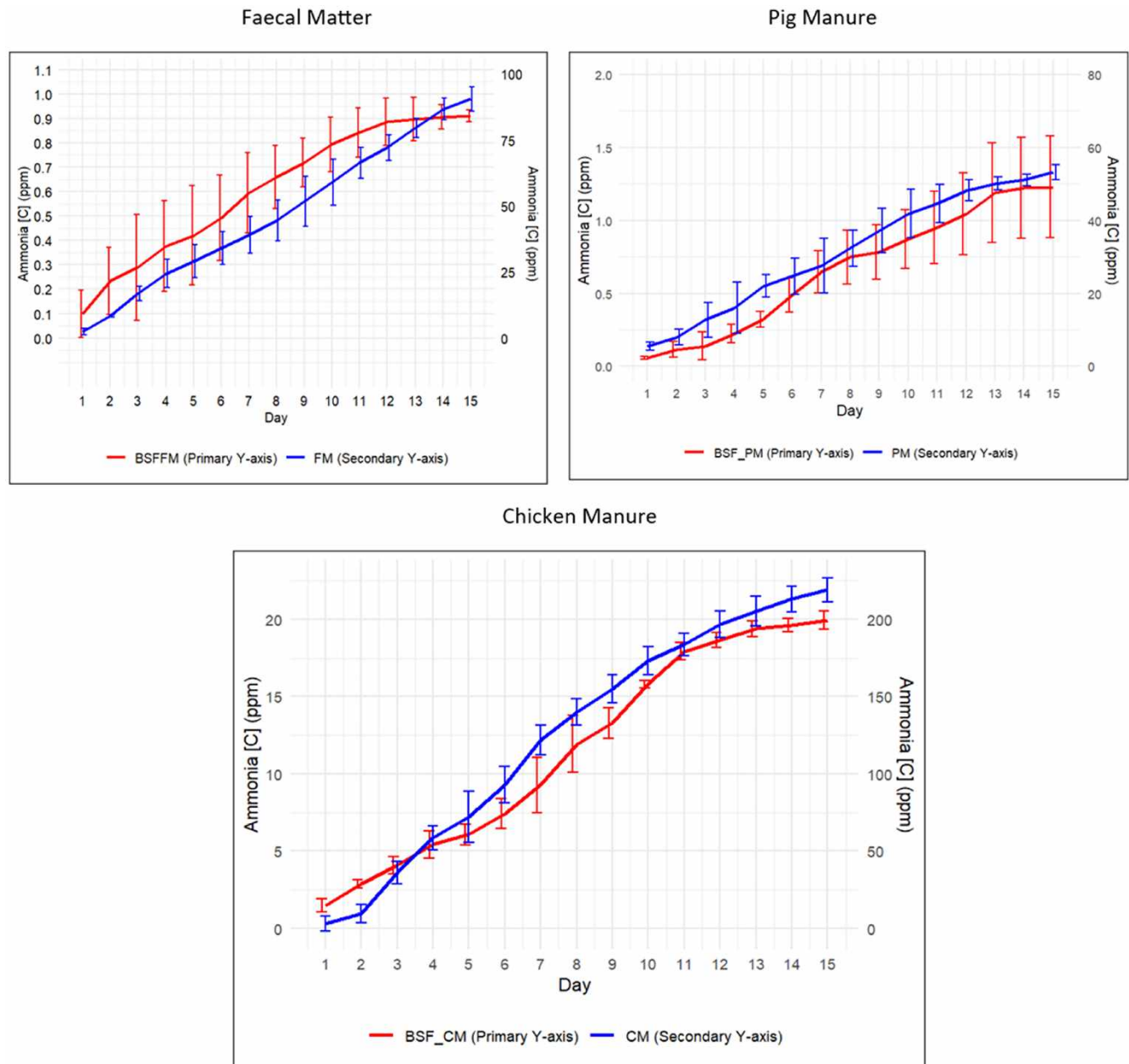


Figure 5 | Cumulative gas concentration for ammonia emitted from FM, PM, and CM with BSF and without BSF.

substrate aeration. CM recorded the highest concentration among the control substrates, followed by FM and PM, reflecting differences in content and substrate composition. The elevated NH_3 level in the control substrates was linked to the decomposition of nitrogen-rich compounds, as shown by elemental analysis (Table 2). The high nitrogen content in FM (5.12%) and PM (2.41%), and the presence of a low C/N ratio in CM (8.51), enhanced ammonia volatilization during microbial degradation. In contrast, BSFL treatment significantly reduced emissions, as the released ammonia was converted and re-assimilated into microbial and larval biomass through ammonium uptake and microbial nitrification.

Previous studies have shown that BSFL can consume 50% of the manure while digesting 62% of the nitrogen element (Rehman *et al.* 2022). A similar study revealed that BSFL altered the microbial community structure, enhancing the presence of beneficial microbes while inhibiting ammonifiers, which further mitigates ammonia emissions (Li *et al.* 2023). Though

under various conditions, untreated manure exhibited high emissions, which were comparable with those recorded in this study, which recorded high emissions in FM. The p -value of 5.76×10^{-7} indicates a highly significant reduction in NH_3 emissions observed in BSFL-treated substrate, demonstrating the effectiveness of larvae in mitigating ammonia release during bioconversion. Similarly, in PM, a p -value of 5.70×10^{-8} further supports a significant reduction of NH_3 emissions under BSFL treatment.

The reduction across BSFL-treated substrates results from the larvae's capacity to assimilate nitrogen into body protein during bioconversion, thereby limiting the nitrogen available for volatilization as ammonia. This process improves nitrogen retention and produces nutrient-rich frass suitable for use as organic fertilizer. Previous studies (Chen *et al.* 2019; Lopes *et al.* 2022; Parodi *et al.* 2022) further showed that BSFL bioconversion promotes nitrogen stabilization and reduces NH_3 losses through biological transformation into stable organic forms, supporting the effectiveness of BSFL in mitigating nitrogen-related emissions during waste treatment. Overall, these results reaffirm that BSFL bioconversion offers a dual advantage by minimizing nitrogen-related gaseous emissions while enhancing nutrient recovery, making it a promising approach for sustainable waste management.

Nitrous oxide emissions on the three substrates

The N_2O emissions from the tested substrates are presented in Figure 6.

The N_2O emissions varied significantly across the substrates. All BSFL treatments resulted in a decline in cumulative N_2O emissions compared to the control groups. The most significant reduction occurred in CM, where emissions dropped from $44,294.91 \pm 200.00$ ppm to 0.83 ± 0.01 ppm, representing over a 99% decrease. FM recorded a sharp decline from $3,619.55 \pm 100.00$ ppm to 0.10 ± 0.11 ppm, while PM exhibited relatively low emissions, decreased from 0.55 ± 0.01 ppm to 0.16 ± 0.60 ppm. The statistically significant p -values (2.56×10^{-6} , 3.84×10^{-9} , and 4.11×10^{-7}) further confirmed reduced N_2O emissions across all BSFL-treated substrates. The observed trend results from the larvae's capacity to assimilate nitrogen into their biomass, thereby reducing the pool of inorganic nitrogen available for nitrification and denitrifying the principal microbial pathway leading to N_2O .

The relatively low N_2O emissions can be linked to the elemental composition of the substrates. FM, with the highest nitrogen content (5.12%), recorded slightly higher N_2O concentration, while overall emissions remained low due to limited available nitrogen. However, BSFL treatment minimized N_2O release by altering substrate conditions that typically promoted denitrification. The larval activity enhanced oxygen diffusion and anaerobic microsites, limiting the conversion of nitrate to N_2O . Additionally, nitrogen was immobilized in stable ammonium forms, thereby restricting substrate available for microbial denitrification (Pang *et al.* 2020). As a result, nitrogen gases are minimized while the remaining frass retains a higher nutrient value. These mechanisms collectively explain the lower N_2O emissions and highlight BSFL bioconversion as an efficient nitrogen stabilizing process (Mertenat *et al.* 2019; Zhang *et al.* 2024). These outcomes position BSFL treatment as a sustainable waste strategy that simultaneously mitigates emissions and enhances nutrient recovery, contributing to circular bioeconomy goals.

Among the substrates, BSFL treatment achieved the highest CH_4 mitigation in FM (91.13%), while CO_2 and N_2O reductions were greatest in CM (99.85 and 99%, respectively); similarly, ammonia showed its highest reduction in FM (98.89%). Overall, these results demonstrate the mitigation capacity of BSFL bioconversion, with its effectiveness varying across gases and substrates depending on the waste characteristics and microbial dynamics. To further quantify the climate implications of these emissions reductions, an LCA was conducted to translate gaseous concentrations into equivalent GWP values.

LCA for BSFL from the bioconversion process

This study focused on GWP, measured in kg CO_2 -equivalent, to assess which substrates, FM, PM, and CM, contributed the most to GHG emissions during the bioconversion process. A summary of the computations is presented in Table 4.

The GWP of each gas determined using openLCA provided insight into the potential contribution of CH_4 , CO_2 , N_2O , and NH_3 emissions from each substrate to climate change, as illustrated in Figure 7.

The GWP results demonstrated that BSFL has significant potential to mitigate GHG emissions, achieving reductions of 94.3% in CM, 91.7% in FM, and 80% in PM. The substrate with the highest GWP cumulatively over the 15 days was CM with emissions of (0.194 kg CO_2 -eq), followed by FM (0.072 kg CO_2 -eq) and PM (0.005 kg CO_2 -eq). The high GWP from CM and FM resulted from rapid microbial degradation and intensive organic matter mineralization, which enhanced

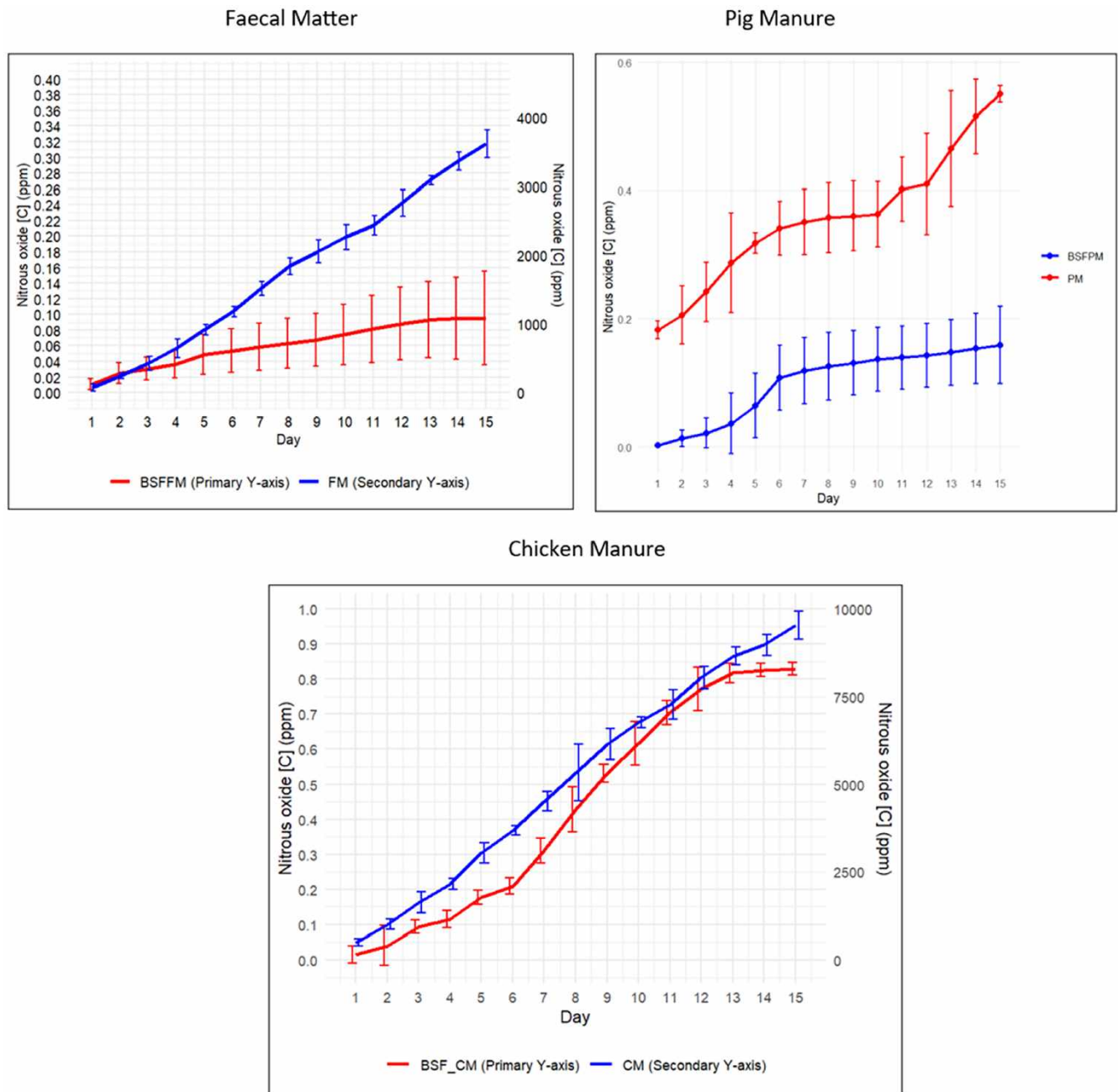


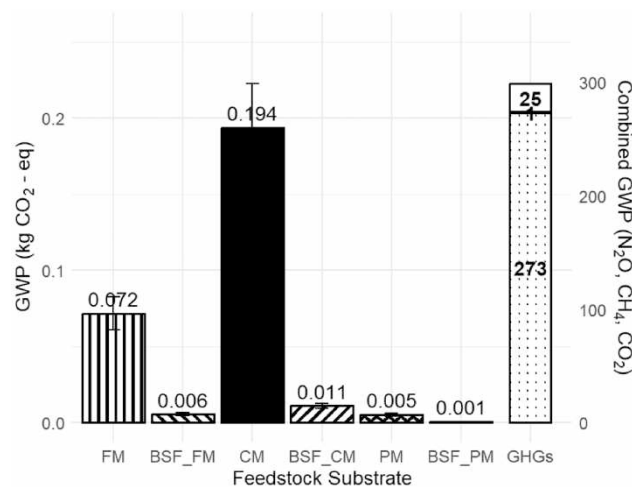
Figure 6 | Cumulative gas concentration for nitrous oxide emitted from FM, PM, and CM with BSF and without BSF.

CO_2 , CH_4 , and N_2O emissions during composition (Bolu *et al.* 2018; Kacprzak *et al.* 2023; Lambiasi *et al.* 2024). PM produced a lower GWP due to its slower biodegradation and limited nitrogen turnover, conditions that restrict gas generation. Previous studies have confirmed that variations in substrate composition and microbial activity dynamics largely determine the GWP of the untreated wastes (Van der Weerden *et al.* 2021). The high emissions observed here are consistent with these findings, indicating that untreated CM presents serious environmental problems by contributing substantially to climate change.

The treated substrate showed a relatively lower GWP than the control substrate because the larvae's bioconversion process alters the decomposition pathway. The larvae consumed and aerated the organic material, suppressing anaerobic microbial activity and limiting conditions conducive to methane and nitrous oxide generation. Additionally, a significant

Table 4 | Input and outputs of LCA

Category	Inputs					Outputs (cumulative gas concentration over 15 days)				
	Fuel for transportation (L)	Substrate (kg)	Water (L)	BSF eggs (kg)	Chicken feeds (kg)	Frass (kg)	CH ₄ (kg)	CO ₂ (kg)	NH ₃ (kg)	N ₂ O (kg)
FM	3	0.25	0.4			5.25×10^{-2}	2.21×10^{-3}	1.02×10^{-2}	8.76×10^{-5}	3.62×10^{-3}
BSFFM	3	4.25	4	0.01	2	3.99×10^{-1}	1.96×10^{-4}	1.95×10^{-4}	9.71×10^{-7}	9.53×10^{-8}
PM	0.5	0.25	0.4			6.60×10^{-2}	1.49×10^{-4}	1.18×10^{-3}	5.33×10^{-5}	5.50×10^{-7}
BSFPM	0.5	0.25	4	0.01	2	6.47×10^{-2}	2.03×10^{-5}	1.26×10^{-4}	1.22×10^{-5}	1.59×10^{-7}
CM	0.5	0.25	0.4			7.31×10^{-2}	3.54×10^{-3}	9.49×10^{-2}	2.19×10^{-4}	4.43×10^{-2}
BSFCM	0.5	0.25	4	0.01	2	6.50×10^{-2}	3.94×10^{-4}	1.38×10^{-4}	2.00×10^{-5}	9.53×10^{-8}

**Figure 7** | Global warming potential on FM (faecal matter), BSF_FM (faecal matter with black soldier fly), CM (chicken manure), BSF_CM (chicken manure with black soldier fly), PM (pig manure), and BSF_PM (pig manure with black soldier fly), and GHGs (global warming potential for N₂O, CH₄, and CO₂).

portion of the substrate carbon was assimilated into larval biomass rather than being mineralized into CO₂, thereby reducing the overall carbon footprint of the process. This biological stabilization process altered the normal decomposition pathway, thereby preventing the release of GHG typically produced during conventional waste degradation. The same perspective is shared by previous studies, which reported significant GWP reductions in BSFL-treated systems (Octaviani *et al.* 2024).

An LCA's impact scores are determined by the part of the system boundary. When all processes and direct emissions are taken into account, the resulting impact on the functional unit tends to be higher as reported in the previous studies (Boakye-Yiadom *et al.* 2022). For instance, in a study by Smetana *et al.* (2019), the authors included the materials and energy associated with other phases, such as the construction of the processing facility, the production of larvae, and additional frass processing to obtain the final dried organic product of a very high impact of 877 kg CO₂-equivalent/ton dried organic fertilizer. However, Komakech *et al.* (2015) reported comparatively low GWP of 12.40 and 17.36 kg CO₂-eq./ton FU, respectively, mostly due to the exclusion of several other impacting categories from the system boundary, such as substrate collection, transport, pre-treatment, and larval generation.

The present study emissions differed from those of other researchers because of the distinct functional unit and the defined system boundaries used. A functional unit of 0.25 kg was used to accurately represent the laboratory experimental setup and ensure precision in quantifying emissions under controlled conditions (Ichwan *et al.* 2021). This means that the emissions were calculated based on the treatment of exactly 0.25 kg of each type of waste, ensuring a standardized basis for assessing environmental impact. Fuel consumption during transportation was automatically incorporated as an input parameter within the OpenCLA software. Several processes, such as electricity consumption, infrastructure development, chick

mash production, and frass collection, were excluded from the system boundary. Although MQ-series sensors are subjected to cross-sensitivity and $\pm X\%$ uncertainty, this uncertainty affected all treatments proportionally. Since GWP calculations are based on relative differences between control and treated substrates under identical monitoring conditions, the comparative mitigation efficiency remained robust, although absolute emission magnitudes may carry systematic uncertainty.

CONCLUSION

This study quantified GHG emissions from CM, PM, and human FM, and evaluated the mitigation potential of BSFL bioconversion. The results clearly demonstrated that untreated substrates emitted substantially higher concentrations of methane (CH_4), CO_2 , nitrous oxide (N_2O), and ammonia (NH_3) compared to substrates treated with BSFL. Among the untreated substrates, CM produced the highest cumulative emissions and GWP, followed by human FM and PM. BSFL treatment significantly reduced GHG emissions across all substrates. The corresponding reduction in GWP was 94.3% for CM, 91.7% for human FM, and 80% for PM. The highest mitigation efficiency was observed in CM, indicating that nitrogen-rich substrates with low carbon-to-nitrogen ratios benefit most from larval bioconversion. These findings confirm that substrate composition strongly influences emission dynamics and the overall climate mitigation performance of BSFL treatment.

The study addressed the key research question of whether BSFL bioconversion can effectively reduce GHG emissions from different organic waste substrates. The results provide clear evidence that larval activity reduces emissions by enhancing substrate aeration, assimilating carbon and nitrogen into biomass, and suppressing microbial pathways responsible for methane and nitrous oxide production.

Although conducted under controlled laboratory conditions, the findings provide a strong experimental basis for considering BSFL bioconversion as a low-emission alternative to conventional manure management practices. Future research should focus on validating these results under field-scale conditions and assessing long-term operational performance. Overall, BSFL bioconversion demonstrates significant potential as a sustainable strategy for reducing GHG emissions from organic waste streams.

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CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

V.W.G.: conceptualization, writing – original draft, writing literature review and editing, designing the methodology, data collection, data curation, and analysis. J.M.R.: Conceptualization, writing – review & editing, methodology, validation, supervision, funding acquisition, and project administration. E.K.N.: Conceptualization, writing – review & editing, methodology, validation, and supervision. R.M.M.: Conceptualization, writing – reviews and editing, data curation, supervision, funding acquisition, project administration, methodology review, and manuscript writing. All authors reviewed and approved the final version of the manuscript.

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ETHICS STATEMENT

All substrates were conducted with the farmers' oral consent and in accordance with institutional guidelines for safe ethical research.

DATA AVAILABILITY STATEMENT

All dataset used for supporting the conclusion of this article are available from the corresponding authors upon request.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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